



$$R_{DC} = \dots\dots\dots \Omega$$

$$R_1 = \frac{3}{2} \times 1.2 \times R_{DC} = \dots\dots\dots \Omega$$

$$R_{01} = \frac{W_{sc}}{3I_s^2} = \dots\dots\dots \Omega$$

$$R'_2 = R_{01} - R_1 = \dots\dots\dots \Omega$$

$$\frac{BG}{BD} = \frac{R'_2}{R_{01}}; \quad BG = \dots\dots\dots \times BD$$

Selection of current and power scale

$$\text{Current scale} = 1 \text{ cm} = \dots\dots\dots \text{A}$$

$$I_0 = \dots\dots\dots \text{A} (= \dots\dots\dots \text{cm})$$

$$I_{sN} = I_{sc} \left(\frac{V_{rated}}{V_{sc}} \right) = \dots\dots\dots \text{A} (= \dots\dots\dots \text{cm})$$

$$\phi_0 = \cos^{-1} \left(\frac{W_{oc}}{\sqrt{3} V_{oc} I_{oc}} \right) = \dots\dots\dots$$

$$\phi_{sc} = \cos^{-1} \left(\frac{W_{sc}}{\sqrt{3} V_{sc} I_{sc}} \right) = \dots\dots\dots$$

$$\text{Power Scale} = \sqrt{3} \times V_{rated} \times \text{current scale} = \dots\dots\dots \text{W} = 1 \text{ cm}$$

PERFORMANCE AT FULL LOAD FROM CIRCLE DIAGRAM

$$\text{Full load output} = \dots\dots\dots \text{W} = PQ = \dots\dots\dots \text{cm}$$

$$\text{Full load current} = OP \times \text{current scale} = \dots\dots\dots \times \dots\dots\dots = \dots\dots\dots \text{A}$$

$$\text{Full load power factor} = \frac{PT}{OP} = \dots\dots\dots \text{lag}$$

$$\text{Rotor copper loss at full load} = QR \times \text{power scale} = \dots\dots\dots \times \dots\dots\dots = \dots\dots\dots \text{W}$$

$$\text{Stator copper loss at full load} = RS \times \text{power scale} = \dots\dots\dots \times \dots\dots\dots = \dots\dots\dots \text{W}$$

$$\text{Constant loss} = ST \times \text{power scale} = \dots\dots\dots \times \dots\dots\dots = \dots\dots\dots \text{W}$$

$$\text{Rotor input at full load} = PR \times \text{power scale} = \dots\dots\dots \times \dots\dots\dots = \dots\dots\dots \text{W}$$

$$\begin{aligned} \text{Torque at full load} &= PR \times \text{power scale (sync. watts)} = PR \times \text{power scale} \times \frac{60}{2\pi N_s} \text{ N-m} \\ &= \dots\dots\dots \times \dots\dots\dots \times \frac{60}{2\pi \times N_s} = \dots\dots\dots \text{N-m} \end{aligned}$$

$$\text{Motor input at full load} = PT \times \text{power scale} = \dots\dots\dots \times \dots\dots\dots = \dots\dots\dots \text{W}$$

$$\text{Efficiency at full load} = \frac{PQ}{PT} \times 100\% = \dots\dots\dots \%$$

$$\text{Slip at full load, } s = \frac{QR}{PR} \times 100\% = \dots\dots\dots \%$$

$$\text{Speed at full load} = (1-s) \times N_s = \dots\dots\dots \text{rpm}$$